

Brāhmasphuṭsiddhānta

of Brahmagupta (628 CE)

Critical Edition with English Commentary

Edited by

Sudhākara Dvivedin

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Part I, Chunk 3

Pages 281–420 of the Original Edition

This chunk comprises:

*Chapters VIII–XIII of the English Commentary,
and the opening Sanskrit text of the Madhyamādhikāra*

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Introduction to This Chunk

This volume contains pages 281–420 of the critical edition of Brahmagupta’s *Brāhmasphuṭsiddhānta*, edited by Sudhākara Dvivedin. The content falls into two major divisions:

1. **English Commentary and Analysis (Chapters VIII–XIII):** A detailed scholarly exposition of Brahmagupta’s contributions to algebra, astronomy, and astronomical instruments, with extensive mathematical commentary.
2. **Sanskrit Text:** The critically edited Sanskrit text of the *Brāhmasphuṭsiddhānta*, beginning with the first chapter (Madhyamādhikāra), accompanied by a critical apparatus recording variant readings.

The English commentary covers Brahmagupta’s work on the kuṭṭaka (pulverizer) method for solving linear Diophantine equations, the varga-prakṛti (square-nature, or Pell’s equation), rational geometrical figures, astronomical computations including the divisions of the zodiac, and descriptions of astronomical instruments.

1 Chapter VIII: Brahmagupta as an Algebraist

This chapter (original pages 232–277) presents Brahmagupta’s algebraic methods, particularly the kuṭṭaka (pulverizer) operations for solving indeterminate equations of the first and second degrees. The commentary draws extensively from the works of later mathematicians including Bhāskara I, Prthūdaka Svāmi, and others.

1.1 The Pulverizer (Kuṭṭaka)

Brahmagupta’s kuṭṭaka method is a systematic procedure for solving linear Diophantine equations of the form:

$$ax \pm c = by$$

where a , b , and c are given integers, and x , y are the unknowns to be found in integers.

The text says that as a preliminary operation to the solution of this pulveriser, a and b , i.e., civil days in *yuga* and revolution-number of the planet, should be made prime to each other by dividing them out by their greatest common factor. That is to say, in solving a pulveriser, one should always make use of abraded divisor and abraded dividend.

The interpolator, i.e., the residue, should also be divided out by the same factor. (This instruction is not given in the text, but it is implied that the residue should be computed for the abraded dividend and abraded divisor.)

The procedure is described as follows:

Set down the dividend above and the divisor (*hāra*) below that. Divide them mutually and write down the quotients (*labdha*) of division one below the other (in the form of a chain). (When an even number of quotients is obtained) think out by what number the (last) remainder be multiplied so that the product being diminished by the (given) residue be exactly divisible (by the divisor corresponding to that remainder). Put down the chosen number called *mati* below the chain and then the new quotient underneath it. Then by the chosen number multiply the number which stands just above it, and to the product add the quotient (written below the chosen number). (Replace the upper number by the resulting sum and cancel the number below). Proceed afterwards also in the same way

(until only two numbers remain). Divide the upper number (called the “multiplier”) by the divisor by the usual process and the lower one (called the “quotient”) by the dividend: the remainders (thus obtained) will respectively be the *ahargana* and the revolutions etc. or what one wants to know.

In modern notation, if $x = \text{ahargana}$ and $y = \text{complete revolutions performed by the planet}$, the pulverizer finds integer solutions to the congruence arising from planetary motion calculations.

1.1.1 Sanskrit Text of the Kuṭṭaka Rule

Brahmagupta states the pulverizer rule in the following verses (extitBrSpSi. XVIII. 32–33):

छेदाभ्यामपवर्त्य हारभाजकौ छेदसम्मूर्च्छितेन च क्षेपेण । हारेण हृतशेषं भाजकहतं च योगतो वा क्षेपं संयोज्य वियोज्य
वा ॥ ३२ ॥
छिन्नं छेदेन युक्तं वा शुद्धिकृतं विभाजितं पुनः । लब्धिरुपरि निहिता सा विपर्यस्ता यथोत्क्रमेण ॥ ३३ ॥

extbfTranslation: “Having reduced the dividend and divisor by their common factor, and the interpolator also by that same factor; the residue (of the dividend), multiplied by the interpolator and added to or subtracted from the quotient, is divided by the reduced divisor. The quotient is placed above (in the chain), reversed in order (from the last to the first).”

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1.2 Linear Equations

Proceeding as before, we find solutions to equations such as:

$$11y = 18x + 10$$

to be $x = 8, y = 14$. These will also be the values of x and y in the case of the negative divisor but the quotient for the reasons stated before should be made negative. So the solution of:

$$-11y = 18x + 10$$

is $x = 8, y = -14$. Subtracting these (i.e., their numerical values) from their respective abraders, we get the solution of:

$$-11y = 18x - 10$$

as $x = 3, y = -4$.

Bhāskara I’s rule states:

“When the divisor is positive or negative the numerical values of the quotient and multiplier remain the same: when either the divisor or the dividend is negative, the quotient must always be known to be negative.”

1.2.1 Brahmagupta’s Rule on Signs in Equations

Brahmagupta gives the following rule regarding the signs of solutions (extitBrSpSi. XVIII. 30):

धनर्णयोर्भाजकहारयोस्तुल्यं लब्धं गुणकमपि तुल्यं । भाजकधनर्णे विरले लब्धं ऋणमेवं ज्ञेयं सदा ॥ ३० ॥

extbfTranslation: “When the divisor and dividend are both positive or both negative, the quotient and multiplier are (numerically) the same. When the divisor is positive and negative (respectively), the quotient must always be known to be negative.”

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1.3 One Linear Equation in More Than Two Unknowns

Whenever a linear equation involves more than two unknowns the Indian algebraists used to assume arbitrary values for all the unknowns except two and then to apply the method of *kuṭṭaka* or “pulveriser.” In this connection, Brahmagupta says:

The method of the pulveriser (should be employed if there be present many unknowns in any equation).

1.4 Varga-Prakṛti: The Square-Nature (Pell’s Equation)

The varga-prakṛti is one of the most significant contributions of Indian algebra. The problem is to solve, in integers, equations of the form:

$$Nx^2 \pm c = y^2$$

where N is a given non-square integer, and x, y are unknown integers. When $c = 1$, this is Pell’s equation.

1.4.1 First Lemma of Brahmagupta

If (x_1, y_1) and (x_2, y_2) be two solutions of:

$$Nx^2 + k_1 = y^2 \quad \text{and} \quad Nx^2 + k_2 = y^2$$

respectively, then a new solution of:

$$Nx^2 + k_1k_2 = y^2$$

is given by:

$$x = x_1y_2 \pm x_2y_1, \quad y = y_1y_2 \pm Nx_1x_2$$

This is known as the *prakṛti-prakṛti* or “composition” principle.

1.4.2 Sanskrit Text of the Composition Principle

Brahmagupta expresses the composition (extitprakṛti-prakṛti) principle as follows (extitBrSpSi. XVIII. 64–65):

यावद् वपुः प्रकृतिहृतं द्विघ्नं युग्मं यदि पदं तु तदेव । जातं पुनः प्रकृतिहृतं समसंज्ञं ततस्तत्संयुक्तं ॥ ६४ ॥

अन्योन्यहतं युगपदेतयोः पदाभ्यां त्रैराशिकेन । क्षेपक्षेपिणोर्वा स्यातां पदे क्षेपयुतेऽथवा ॥ ६५ ॥

extbfTranslation: “The product of the two (previous) roots is multiplied by the extitprakṛti (the given non-square number); the sum of the cross-products gives the new first root. The product of the two (previous) roots added to the product of their extitprakṛti-multiplied counterparts gives the second root; or alternatively, the sum or difference of the extitkṣepa products gives the new interpolator.”

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1.4.3 Second Lemma of Brahmagupta

Suppose the varga-prakṛti (Square-nature) to be:

$$Nx^2 \pm p^2d = y^2$$

so that its interpolator (*kṣepa*) p^2d is exactly divisible by the square p^2 . Then, putting therein $u = x/p$, $v = y/p$, we derive the equation:

$$Nu^2 \pm d = v^2$$

whose interpolator is equal to that of the original Square-nature divided by p^2 . It is clear that the roots of the original equation are p times those of the derived equation.

1.4.4 Sanskrit Text of the Interpolator Reduction

Brahmagupta gives the rule for reducing the interpolator (extitBrSpSi. XVIII. 66):

छेदः क्षेपस्य तत्पदं भाज्यं तत्पदहतं विभाज्यं । लब्धौ तत्र तदेकराशिः क्षेपेण तुल्यौ तौ ज्ञेयौ ॥ ६६ ॥

extbfTranslation: “The interpolator is divided by the square of any number; its root is the divisor. The (first) root is divided by that number, and the second root is also divided by that number. The two quotients thus obtained are the roots corresponding to the reduced interpolator.”

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1.4.5 Rational Solution

Indian algebraists have usually suggested the following method to obtain a first solution of $Nx^2 + 1 = y^2$:

Take an arbitrary small rational number, α , such that its square multiplied by the *gunaka* N and increased or diminished by a suitably chosen rational number k will be an exact square:

$$N\alpha^2 \pm k = \beta^2$$

where α , k , and β are rational numbers. Let us call this relation the *Auxiliary Equation*. Then by Brahmagupta’s Corollary, we get:

$$N(2\alpha\beta)^2 + k^2 = (\beta^2 + N\alpha^2)^2$$

or:

$$N \left(\frac{2\alpha\beta}{k} \right)^2 + 1 = \left(\frac{\beta^2 + N\alpha^2}{k} \right)^2$$

Hence, one rational solution is:

$$x = \frac{2\alpha\beta}{k}, \quad y = \frac{\beta^2 + N\alpha^2}{k}$$

1.4.6 Sanskrit Text of the Rational Auxiliary Method

Brahmagupta describes the method of finding an initial rational solution (extitBrSpSi. XVIII. 67):

इष्टं यदा तदा येन हतं पदं युतं हृतं वा । वर्गः स्यात् ततः क्षेपः प्रकृतिकृतो विद्यते यः सः ॥ ६७ ॥

extbfTranslation: “An arbitrary number is chosen; when multiplied by the extitprakṛti and increased or diminished by some suitable interpolator, it becomes a perfect square. From this, the auxiliary equation is formed and the roots are derived by composition.”

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1.4.7 The Cakravāla or Cyclic Method

The *cakravāla* (cyclic method) is an iterative algorithm for obtaining integer solutions to $Nx^2 + 1 = y^2$. Starting from an auxiliary equation $Na^2 + k = b^2$, the method proceeds by the principle of composition to reduce the interpolator k successively until it becomes ± 1 , ± 2 , or ± 4 , from which the solution with interpolator $+1$ can be derived by known lemmas.

1.4.8 Sanskrit Text of the Cyclic Method

The extitcakravāla method is described by Brahmagupta (extitBrSpSi. XVIII. 68–69):

आभ्यां युक्तं व्यस्तं तावन्मूलयोः क्षेपमूलयोर्युतं । तस्मात् प्रकृतिहृतं लब्धं सूक्ष्मं स्यात् ततः पुनः ॥ ६८ ॥
क्षेपः सूक्ष्मो यावत् तावदेतद् विधानमाश्रितं । यदा महान् तदा तु मूलं विवरेण वर्जितं ॥ ६९ ॥

extbfTranslation: “By the composition of the two roots with the sum or difference of their interpolator-roots, a new root is obtained. The extitprakṛti divided by this gives a small interpolator. This process is repeated until the interpolator becomes small (± 1 , ± 2 , or ± 4). When the interpolator is large, the root is diminished by an assumed number.”

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1.5 Rational Geometrical Figures

Brahmagupta in connection with the solution of rational triangles says:

The square of the optional (*iṣṭa*) side is divided and then diminished by an optional number; half the result is the upright, and that increased by the optional number gives the hypotenuse of a rectangle.

If m, n be any two rational numbers, then the sides of a right-angled triangle will be:

$$m, \quad \frac{1}{2} \left(\frac{m^2}{n} - n \right), \quad \frac{1}{2} \left(\frac{m^2}{n} + n \right)$$

Brahmagupta was the first to give a solution of the equation $x^2 + y^2 = z^2$ in integers. His solution is:

$$m^2 - n^2, \quad 2mn, \quad m^2 + n^2$$

m and n being two unequal integers.

1.5.1 Sanskrit Text on Right-Angled Triangles

Brahmagupta gives the following rule for generating right-angled triangles (*BrSpSi*. XII. 33):

वर्गोऽर्धितो राश्योर्योऽन्तरेण हतो राश्यन्तरेण वर्जितः । कोटिः स भवति सा युक्ता तस्यैव त्रिज्या भवेत् ॥ ३३ ॥

Translation: “The square of the greater number is halved and divided by the difference of the numbers; diminished by the difference, it becomes the upright (*koti*); that (same quotient) increased by the difference becomes the hypotenuse (*trijyā*).”

Thus if $m = 7$ and $n = 4$, then $m^2 - n^2 = 33$, $2mn = 56$ and $m^2 + n^2 = 65$; then the three numbers 33, 56 and 65 bear the relation $33^2 + 56^2 = 65^2$.

1.5.2 Isosceles Triangles with Integral Sides

The following statement of Brahmagupta in this connection is very significant (*BrSpSi*. XII. 35):

The sum of the squares of two unequal numbers is divided by their difference; the quotient is multiplied by the difference and increased by the quotient; the moieties of the sum and difference are the sides of an isosceles triangle.

1.5.3 Sanskrit Text on Isosceles Triangles

The rule for isosceles triangles with rational sides (*BrSpSi*. XII. 35):

यस् तयोरसमयोर्वर्गयोगो भाजितोऽन्तरेण स राशिः । स हतोऽन्तरयुतो द्विधा समद्विभुजस्य भुजे ते ॥ ३५ ॥

Translation: “The sum of the squares of two unequal numbers is divided by their difference; the quotient is multiplied by the difference and increased by the quotient; the moieties of these are the equal sides of an isosceles triangle.”

1.5.4 Rational Inscribed Quadrilaterals

Brahmagupta gave the remarkable formula for the area of a cyclic quadrilateral:

$$A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$$

where $s = \frac{1}{2}(a+b+c+d)$ is the semiperimeter. This generalizes Heron’s formula for triangles (the case $d = 0$).

1.5.5 Sanskrit Text on the Area of a Cyclic Quadrilateral

Brahmagupta’s celebrated formula for the area of a cyclic quadrilateral (*BrSpSi*. XII. 21):

समपरतलभुजयोर्विषमयोः सम्प्रयुक्तयोः सम्मितयोः । विषमयोर्वर्गयुतेः स्यातां द्वे समकोट्यौ द्वे च ॥ २१ ॥

And the rule for the exact area (*BrSpSi*. XII. 28):

सर्वदोर्युतीर्द्धिघ्ना विषमाणि तान्यथान्तराणि यानि । तेषां वर्गेस्तुल्यः क्षेत्रफलवर्ग उद्दिष्टः ॥ २८ ॥

Translation: “The sides of a quadrilateral with equal and unequal sides are given. Twice the sum of all sides, diminished by each side separately; the square root of the product of the remainders gives the exact area.”

1.6 Double First Degree Equations

Brahmagupta’s own rule for solving simultaneous equations is illustrated by the following example. On subtracting from the product of signs and degrees of the Sun, three and four times (respectively) those quantities, ninety is obtained. Determining the Sun within a year (one can pass as a proficient) mathematician.

If we presume x to denote the signs and y the degrees of the Sun, then the equation would be:

$$xy - 3x - 4y = 90$$

Prthūdaka Svāmi solves it in two ways by assuming arbitrary values for one unknown and solving for the other.

1.6.1 Sanskrit Text on Simultaneous Equations

Brahmagupta’s rule for solving simultaneous equations (*BrSpSi*. XVIII. 43):

यावत्तावद् द्वयोरेकस्मिन् मूले क्षिप्ते राश्योर्- अन्यतरराशौ यावत्तावद्विशोध्यावशिष्टं ॥ ४३ ॥

Translation: “Of two unknown quantities, one is assumed arbitrarily; the other is determined by the method of the pulverizer. Having assumed a value for one unknown, substitute it into the equation and solve for the other by kuṭṭaka.”

2 Chapter X: Arabic and Indian Divisions of the Zodiac

This chapter (original pages 278–289) discusses the correspondence between the Indian system of 27 or 28 nakṣatras (lunar mansions) and the Arabic and Greek divisions of the zodiac.

The author, following Colebrooke’s researches, enumerates the twenty-seven (or twenty-eight) nakṣatras and their principal stars, comparing them with the Arabic *manāzil al-qamar* (moon’s mansions). The chapter begins with Aśvinī and proceeds through the complete cycle.

2.1 Nakṣatras and Their Principal Stars

The principal nakṣatras discussed include:

1. *Aśvinī* — The Wives of the Aśvins, figured as a horse’s head, comprising three stars. The principal star is identified as β Arietis (Sheratan).
2. *Bharaṇī* — Figured as the body of a horse, containing three stars, of which 35, 39, and 41 Arietis are the brightest.
3. *Kṛttikā* — The Pleiades, containing six (or seven) stars.
4. *Rohiṇī* — The Red One, identified with Aldebaran (α Tauri).
5. *Mṛgasīras* — The deer’s head, identified with λ Orionis.

2.1.1 Sanskrit Names of the Nakṣatras

Brahmagupta enumerates the twenty-seven nakṣatras in the following verse (*BrSpSi*. IX. 1–2):

अश्विनी भरणी कृत्तिका रोहिणी मृगशीर्ष च । आर्द्रा पुनर्वसुः पुष्य आश्लेषा मघा पूर्वफल्गुनी ॥ १ ॥
उत्तरफल्गुनी हस्त चित्रा स्वाती विशाखा चानूराधा । ज्येष्ठा मूलं पूर्वाषाढोत्तराषाढा श्रवणं धनिष्ठा ॥ २ ॥
शतभिषक् पूर्वभाद्रपदोत्तरभाद्रपदं रेवती । एता याम्योत्तराः सप्तविंशतिर्नक्षत्राणि ॥

Translation: “Aśvinī, Bharaṇī, Kṛttikā, Rohiṇī, Mṛgasīras, Ārdrā, Punarvasu, Puṣya, Āślēṣā, Maghā, Pūrvaphalgunī, Uttaraphalgunī, Hasta, Citrā, Svāti, Viśākhā, Anūrādhā, Jyēṣṭhā, Mūla, Pūrvāṣāḍhā, Uttaraṣāḍhā, Śravaṇa, Dhaniṣṭhā, Śatabhiṣak, Pūrvabhādrapadā, Uttarabhādrapadā, and Revatī — these are the twenty-seven nakṣatras from south to north.”

2.1.2 Sanskrit Text on Rāśis (Zodiacal Signs)

Brahmagupta describes the twelve rāśis (*BrSpSi*. IX. 3–4):

मेषवृषमिथुनं कर्कटं सिंहकन्ययुते तुलां च । वृश्चिकं धनुर्मकरं कुम्भं मीनं च राशयो द्वादश ॥ ३ ॥
तुलादयो याम्याः क्रान्तिविशेषास्त एव ते द्वादश । प्राच्य उदययने दक्षिणतः प्रवृत्ताः परिलिखिताः ॥ ४ ॥

Translation: “Meṣa (Aries), Ṽṛṣa (Taurus), Mithuna (Gemini), Karkāṭa (Cancer), Siṃha (Leo), Kanyā (Virgo), Tulā (Libra), Ṽṛścika (Scorpio), Dhanus (Sagittarius), Makara (Capricorn), Kumbha (Aquarius), and Mīna (Pisces) — these are the twelve rāśis. Those beginning with Tulā are southern (having south declination). These same twelve are marked as eastern in the northern course (*udayana*), proceeding southward.”

The chapter continues through all 27 nakṣatras, noting their shapes, number of stars, principal stars, and their Arabic counterparts. Key comparisons include:

- The Arabic *Alnath* (β Tauri) corresponds to the Indian star in Mṛgasīras.
- The division of the ecliptic into 27 or 28 mansions is of purely Indian origin, while the concept of 12 Rāsis (signs) is probably inspired from the Greeks.
- The names Kanyā (Virgo), Tulā (Libra), and Karka (Cancer) appear to be of Greek origin.

The author concludes by noting that while the concept of 27 nakṣatras is Vedic and most ancient, the agreements between Indian and Arabic divisions are largely by chance.

3 Chapter XI: Brahmagupta's Astronomy: Its Highlights

This chapter (original pages 290–319) presents the essential astronomical computations of the Brāhmasphuṭsiddhānta, including the computation of time units, mean longitudes of planets, and various astronomical elements.

3.1 Units of Time

Brahmagupta's system of time reckoning is as follows:

- A *kalpa* consists of 4,320,000,000 solar years.
- The number of civil days in a kalpa is 1,577,917,828,000.
- The number of solar months is 51,840,000,000.
- The number of lunar months is 53,433,300,000.
- By subtracting, the number of *adhi-māsas* (additional months) is 1,593,300,000.
- This multiplied by 30 gives the number of lunar days (*śāsi-divasa*) in a kalpa: 1,602,999,000,000.

Calculation of Ahargana: The method of calculating ahargana (number of days elapsed since the beginning of kaliyuga) involves converting the given date into elapsed days from the epoch.

Brahmagupta calculates the *sṛṣṭi-saṃvatsara* or the Creation Era during his year of composition:

Six *manus* have gone in the kalpa; the seventh manu is now running of which have lapsed 27 caturyugis; of the twenty eighth caturyugi, the three yugas, *kṛta*, *dvāpara* and *tretā* have gone by and also of the present kaliyuga 3179 years have lapsed.

The total period thus lapsed on calculation comes to be 1,972,947,179 years:

$$\begin{aligned}
 \text{Total Period} &= 6 \text{ manus} + 7 \text{ manu-sandhis} + 27 \text{ yugas} \\
 &\quad + \text{kṛta} + \text{dvāpara} + \text{tretā} + 3179 \text{ years of kali} \\
 &= (6 \times 71 \times 4,320,000) + (7 \times 4 \times 432,000) \\
 &\quad + (27 \times 4,320,000) + (1,728,000 + 1,296,000 + 864,000) + 3,179 \\
 &= 1,972,947,179 \text{ years.}
 \end{aligned}$$

3.1.1 Sanskrit Text on Time Reckoning

Brahmagupta's verses defining the divisions of time (*BrSpSi*. I. 5–7):

नाडी त्रिंशता षष्ट्या नक्षत्रदिनेन तुल्यः कालः । दशभिश्च नाडीभिः प्राणः प्राणैर्विनाडी दशभिः ॥ ५ ॥

षड्भिः कलाः पलं च तैः षड्भिर्नाडी तु काष्ठा । दशभिर्नाडिका मुहूर्तः सवनो दिनं त्रिंशता ॥ ६ ॥

मासस्त्रिंशता दिवसानां राश्यष्टकं भवेत् त्रिंशता तु । संवत्सरः सावनः सौरस्तावानेवेन्दवः स्यात् ॥ ७ ॥

Translation: “A nāḍī (time-unit) is equal to thirty (vināḍī)s or sixty (kāḷā)s of a nakṣatra day. Ten nāḍīs make a prāṇa; ten prāṇas make a vināḍī. Six kāḷās make a pala; six palas make a nāḍī or kāṣṭhā; ten nāḍīs make a muhūrta. A civil (savana) day consists of thirty (such units); a month consists of thirty civil days. Eight rāśis (signs) make thirty (days); a savana year is that much; so also is the lunar year.”

3.2 Mean Longitude of Planets

Brahmagupta gives rules for finding the mean longitudes of the planets. The general method involves:

1. Finding the *ahargana* (elapsed days) from the epoch.
2. Multiplying by the revolution-number of the planet and dividing by civil days in a kalpa.
3. The result gives the mean longitude in signs, degrees, minutes, etc.

3.2.1 Sanskrit Text on Mean Longitudes

Brahmagupta's rule for computing the mean longitude of planets (*BrSpSi*. I. 16–17):

मध्यमानीयं निजकल्पभुक्त्या गतदिनैर्गुणितैः क्षेपयुक्तैः । भुज्यक्षभागैः स्वपरिवर्तहृतैर्भवति मध्यं ग्रहमण्डलस्य ॥ १६ ॥

सूर्येन्दुवोर्मध्यमतः स्फुटं भवति तत्कल्पसिद्धं । शेषेभ्यो मध्यात् स्फुटीकरणं कार्यं प्रयत्नेन ॥ १७ ॥

Translation: “The mean (longitude) is obtained from the elapsed days multiplied by the revolution-number (in a kalpa) and increased by the epoch position; divided by the civil days (in a kalpa), the result gives the mean longitude of the planet in signs, degrees, minutes, etc. For the Sun and Moon, the mean longitude is the true (longitude); for the other planets, corrections must be carefully applied to the mean to obtain the true.”

3.3 Concordance of Working Rules

The editor provides a concordance table showing where Brahmagupta's rules appear in other astronomical texts. Key rules include:

1. Rule for finding the mean longitude of the *Sighrocca* of Venus and also giving the additives for the *Sighrocca* of Mercury and Moon: *BrSpSi*. XXV. 36.
2. Rule for finding the mean longitude of Saturn: *BrSpSi*. XXV. 35.
3. Rule for finding the mean longitude of Mars: *BrSpSi*. XXV. 33.
4. Rule for finding the mean longitude of Jupiter: *BrSpSi*. XXX. 35.

5. Rule for finding the distance of a place from the prime meridian: *BrSpSi*. I. 36.
6. Rule for finding the directions: *BrSpSi*. III. 1.
7. Rule for determining the declination, day-radius, earth sine and ascensional difference (for the Sun or a point on the ecliptic): *BrSpSi*. II. 55.

3.3.1 Sanskrit Text on Planetary Corrections

Brahmagupta gives the rules for the *manda* and *śighra* corrections (*BrSpSi*. II. 1–2):

मन्देन मन्दस्फुटभुजाफलं स्फुटीकरणं कुर्यात् । ततः शीघ्रेण शीघ्रस्फुटभुजाफलं ततो वृत्ते ॥ १ ॥

मन्दफलं स्फुटवर्गार्धात् मूलं क्षिप्त्वा दोःकोट्योः । व्यासार्धहतचापं मन्दस्फुटतः स्फुटं ततः स्यात् ॥ २ ॥

Translation: “First, apply the *manda* correction using the *manda* epicycle to obtain the *manda-sphuṭa*; then apply the *śighra* correction using the *śighra* epicycle. Having added the square of the *manda-phala* to half the square of the hypotenuse, taken the square root, and computed the arc (from the Rsine), the true (longitude) is obtained from the *manda-sphuṭa*.”

3.4 Tables of Planetary Constants

The text includes comparative tables of:

Table 1: Mean Diameters of the Planets				
Planet	<i>BrSpSi</i> .	<i>Sūrya-Siddhānta</i>	Āryabhaṭīya	Modern
Sun	32'31"	32'24"	33" approx.	32'2.36"
Moon	32'1" approx.	32'	31'30"	31'8"
Mars	4'46"	2'	1'17"	9'36"
Mercury	6'14"	3'	2'8"	6'68"
Jupiter	7'22"	3'30"	3'12"	3'14.72"
Venus	9'	4'	6'24"	16.8"
Saturn	5'24"	2'30"	1'36"	2'49.5"

4 Chapter XII: Brahmagupta a Great Critic

This chapter (original pages 320–333) discusses Brahmagupta’s critical engagement with the work of Āryabhaṭa and his followers, as well as his relationship with the astronomer Śīṣeṇa.

4.1 Criticism of Āryabhaṭa

Brahmagupta has criticised Āryabhaṭa and his group for their expressions for determining *lambana* (i.e. the difference of the parallaxes, in longitude, of the Sun and the Moon), and the rule for determining the *avanati* or *nati* (i.e. the difference in parallaxes, in latitude of the Sun and Moon).

Lambana is obtained with the help of five Rsines:

- (i) *madhya-jyā* — the Rsine of the zenith distance of the meridian ecliptic point.
- (ii) *udaya-jyā* — the Rsine of the arc of the horizon intervening between the equator and the ecliptic.

(iii) *drk-kṣepa-jyā* — the Rsine of the zenith distance of the central ecliptic point.

(iv) *drg-jyā*

(v) *drg-gati-jyā*

4.1.1 Sanskrit Text on Parallax Corrections

Brahmagupta's rules for *lambana* and *natī* (*BrSpSi*. V. 8–9):

दृक्क्षेपज्या त्रिज्यकृत्या हता दृग्ज्याऽथवाऽऽभ्यां हता । लम्बकेन हता दृष्टिर्नतज्याकोटिजीवया ॥ ८ ॥
दोर्ज्याकोटिज्ययोर्वर्गयोगमूलं व्यासार्धमाहुः । तत्कर्णहृतं भुजाज्यां लब्धं लम्बनमुच्यते ॥ ९ ॥

Translation: “The *drkkṣepa-jyā* is multiplied by the *trijyā* and divided by the *drg-jyā*; or multiplied by the *śāṅku* and divided by the *lambaka* — this gives the *drṣṭi* (parallax in longitude). The square root of the sum of the squares of the *dorjyā* and *kotijyā* is called the *vyāsārdha* (radius); the *bhujā-jyā* divided by this hypotenuse gives the *lambana* (difference of parallaxes in longitude).”

The *udayaajyā* is given by:

$$\text{udayaajyā} = \frac{R \sin L \times R \sin \varepsilon}{R \cos \phi}$$

where L is the longitude of the horizon ecliptic point in the east, ε the obliquity of the ecliptic, and ϕ the latitude of the place.

4.2 Brahmagupta and Śrīṣeṇa

Brahmagupta also directed his criticism against Śrīṣeṇa, another contemporary astronomer. The chapter explores the doctrinal differences between them regarding planetary computations and the interpretation of astronomical observations.

4.2.1 Brahmagupta's Criticism of Other Astronomers

Brahmagupta states his disagreement with earlier astronomers regarding planetary computations (*BrSpSi*. XI. 1):

आर्यभटदायुष्करणं शिष्यैः परिष्कृतं शिष्याणां । शिष्यैः परिष्कृतं नैव युज्यते तस्मात् तदप्ययुक्तं ॥ १ ॥

Translation: “The calculations of longitudes by Āryabhaṭa, as corrected by his pupils, are not consistent even among the pupils themselves; therefore those (corrections) are also incorrect.”

And on the obliquity of the ecliptic (*BrSpSi*. XI. 5):

अयनवृत्तं भ्रमध्यात् त्रिज्यायास्त्रिभागे क्षितेः स्पर्शात् । तस्मात् षडशीतिर्युग्मदक्षिणमयनं विषुवत् ॥ ५ ॥

Translation: “The ecliptic (*ayana-vṛtta*) touches the Earth at a distance of one-third of the radius from the Earth's center; therefore the greatest declination is 24^{circ} (actually 24^{circ} is the obliquity of the ecliptic), southward in the even (signs) and this is the equinox.”

5 Chapter XIII: Brahmagupta and Astronomical Instruments

This chapter (original pages 334–344) describes the astronomical instruments known to Brahmagupta, drawing on descriptions from the Āryabhaṭīya commentaries and the *Sūrya-Siddhānta*.

5.1 Setting Up of the Gnomon (Śaṅku)

Bhāskara I in his commentary on the Āryabhaṭīya tells us that there was a difference of opinion amongst astronomers in his time regarding the shape and size of the gnomon (also called style). Some prescribed:

- A gnomon with one-third in the bottom of a prism on a square base (*caturaśra*), one-third in the middle of the shape of a cow's tail (*go-pucchākāra*), and one-third at the top of the shape of a spear-head (*śulākāra*).
- A square prismoidal (*samacaturaśra*) gnomon.

The followers of Āryabhaṭa I prescribed a broad (*prthu*), massive (*guru*), and large (*dirgha*) cylindrical gnomon, made of excellent timber and free from any hole, scar, or knot on its body.

Brahmagupta describes a gnomon which at the bottom is two *aṅgulas* wide, pointed as a needle, 12 *aṅgulas* in length, and full of holes from the basic circular part to the pointed extremity (*BrSpSi*. XXII. 39).

For testing the level of the ground:

When there is no wind, place a jar (full) of water upon a tripod on the ground which has been made plane by means of eye or thread, and bore a (fine) hole (at the bottom of the jar) so that the water may fall drop by drop.

5.1.1 Sanskrit Text on the Gnomon

Brahmagupta describes the construction and use of the śaṅku (*BrSpSi*. III. 1–3):

शङ्कुः समचतुरस्रो द्व्यङ्गुलो निर्मलोऽस्त्रिगन्धः । द्वादशाङ्गुलविश्वाग्रे शूलाकारस्तु सूक्ष्मः स्यात् ॥ १ ॥

भूमिं समीकृत्य तोयकुम्भं त्रिपादे स्थापयित्वा । छिद्रेण सूक्ष्मेण तोयं पततीन्दुबिम्बवत् ॥ २ ॥

छायाग्रेण वृत्तं कृत्वा त्रिभिस्त्र्यङ्गुलैः समचतुरस्रं । तस्मिन् विन्यस्य शङ्कुं दिशोऽसावनुगृहीतुं ॥ ३ ॥

Translation: “The gnomon should be quadrangular, two *aṅgulas* wide, clean, without blemish or bad odor, twelve *aṅgulas* long, pointed like a spear-head at the top, and very sharp. Having levelled the ground, place a water-jar on a tripod, bore a fine hole at its bottom so that water falls drop by drop like the disc of the Moon. Having drawn a circle with the shadow-tip, mark a square in it with three *aṅgulas* on each side; having placed the gnomon there, one may determine the directions.”

5.2 Gnomon Used for Finding the Directions

The gnomon was used to determine the cardinal directions by observing the shadow at different times of the day. The method involves marking the shadow positions morning and afternoon when the shadows are equal, and then drawing a line between the two shadow tips to determine the east-west direction.

5.3 Golayantra or Armillary Sphere

The *golayantra* (armillary sphere) is described in detail. After rectifying the meridian, if it be wished to observe the Sun and Moon together, the outer secondary of the ecliptic must be made to intersect the ecliptic at the Sun's place for that time; and the solstitial colure must be moved until the place of intersection be opposite to the Sun.

The author of the *Sūrya-Siddhānta* then proceeds to notice the relation of the great circles before mentioned to the horizon, and observes that whatever place be assumed for the apex of the sphere, the middle of the heaven for that place is its horizon. He concludes by showing that the instrument may be made to revolve with regularity by means of a current of water; and hints that the appearance of spontaneous motion may be given by a concealed mechanism, for which quicksilver is to be employed.

5.3.1 Sanskrit Text on the Golayantra

Brahmagupta describes the armillary sphere (*BrSpSi*. XXII. 1–2):

वलयानि महान्त्यष्टौ निर्माय कांस्यादिना दृढानि । यथोक्तं व्यासतः कुर्याद् भगणानां गतिमैच्छत् ॥ १ ॥

द्युज्यावलयमूर्ध्वार्धःस्थितं मध्यमं द्वितीयं । तृतीयं विषुवद् वलयं चतुर्थं याम्योत्तरं स्यात् ॥ २ ॥

Translation: “Having made eight large rings of bronze or other durable material, firmly constructed, one should make them of the diameters as prescribed, if one desires to observe the motions of the planets. The first ring is the prime vertical, placed above and below; the second is the meridian; the third is the equinoctial colure; the fourth is the solstitial colure.”

5.4 How to Observe Places of Stars

The *Sūrya-Siddhānta* hints that the astronomer should frame a sphere and examine the apparent longitude and latitude (*sphuṭavikṣepa* and *sphuṭadhruvaka*). The commentators describe the method of finding the longitude and latitude of a star by aligning the moveable circles of the armillary sphere.

6 The Sanskrit Text: Madhyamādhikāra

From page 390 of the original edition, the critically edited Sanskrit text of the *Brāhmasphuṭsiddhānta* begins. This section presents the text of the first chapter (Madhyamādhikāra), accompanied by the critical apparatus recording variant readings from manuscripts and earlier editions.

The text begins with the benedictory verse and proceeds through the astronomical rules. The critical apparatus uses sigla to indicate manuscript sources and records emendations and corrections.

॥ श्रीगणेशाय नमः ॥

अथ ब्राह्मस्फुटसिद्धान्तस्य

पूर्वद्वशाध्यायां मध्यमाधिकारः

The opening verses of the Madhyamādhikāra (Chapter I) of the *Brāhmasphuṭsiddhānta* are given below with the critical apparatus showing variant readings:

जयति प्रणतसुरासुरैरगीन्द्ररत्नप्रभाच्छुरितपादपङ्कजः । कर्ता जगत्पतिरदित्यतुल्ययानां महादेवः ॥ १ ॥
 ब्रह्मगुप्तो ग्रहगणितं महता कालेन यत् खल्वीमतम् । क्षीणेऽधुनी तद्विपरीतवद्ब्राह्मणैरुपेतं ॥ २ ॥
 ध्रुवबलार्कसुतबुधज्ञ ज्योतिर्विदां प्रीतिषद्भ्रमदो । पोषणार्थं विवृतन्तरव्यक्तं सह प्रबन्धेन सूचितं ॥ ३ ॥
 चैत्र्यादितदेवद्विजनोद्वन्माससर्वेष युगस्य । सृष्टिच्छादो लंकायां समं प्रवृता दिनेकलस्य ॥ ४ ॥
 प्राग्र्योद्दीनाटिकाद्वो द्विद्विर्यष्टिका त्रिनाटिका षड्घ्ना । घटिका षड्घ्ना दिवलो दिवसानां त्रिंशता मासः ॥ ५ ॥

Critical Apparatus (selected variants):

1. श्री नमः | श्री रामाय — Variant readings from different manuscripts. The benedictory verse varies between recensions.
2. पोषणार्थं विवृतन्तरव्यक्तं for (पोषणार्थं विवृतान्तरव्यक्तं)
3. दिन (च) for (दिन) — Minor orthographic variants.
4. उदकस्य for (उदकस्य) — Manuscript spelling variants.
5. समप्रवृता for (समं प्रवृता)

The text continues with the definitions of time units, and then proceeds to the rules for computing the mean positions of the planets. The critical apparatus throughout records the readings of different manuscript sources, indicated by sigla such as (ग) for one manuscript family, (घ) for another, etc.

6.1 Sample Verses with Commentary

Further verses from the chapter include:

परिवर्तः सप्तचतुःषष्टिर सूर्यस्य दिनस्य मितिः । रवि भगणानां मानः सावनदिनस्य कुट्टकैरेष ॥
 रवि भगणार्धबधा द्वादशांशेन भवति रविमासः । भगणान्तरं रविदिनः शशिमासः सूर्यमासेन ॥
 अधिमासाः शशिमासाविन्द्वदिनस्य भवति शशिदिवसाः । शशिसावनदिवसान्तरनिवमानि तिथिः शशाङ्कदिनं ॥

BrSpSi. I. 22–24

These verses define the relationships between various time units:

- A solar year consists of approximately 365 days 15 minutes 30 seconds.
- A solar month is one-twelfth of the solar year.
- The difference between lunar and solar months gives the *adhi-māsa* (intercalary month).
- The difference between lunar and civil days gives the *tithi* (lunar day).

6.1.1 Verses on Planetary Mean Motions

Further verses giving the revolution-numbers of planets (BrSpSi. I. 12–15):

सूर्यस्य त्र्यक्षरैर्बाणैः सप्तचतुःषष्टिरक्षयुगं भुजगैः । भानोर्द्विषष्टिरष्टादशैः सार्धं शशिनश्चतुर्भिः ॥ १२ ॥
 बुधस्य चतुर्भिर्भानोर्नवभिः सप्तभिर्गुरोः कवेः । भानोर्द्वादशभिः सप्तभिः सितस्य च तारागणस्य ॥ १३ ॥
 भानोः पञ्चभिरष्टभिर्बुधज्ञस्य त्रिभिरक्षयुगं भानोः । एभिः खखेभभगणैर्मनानि ग्रहाणां युगे सूर्यदिनानि ॥ १४ ॥
 चरमं यत् सूर्यदिनं तद् ध्रुवसञ्ज्ञं भवति निर्दिष्टं । तस्मात् प्रवृत्तिः सा स्याद् ग्रहाणां युगपरिवर्त्तनां ॥ १५ ॥

Translation: “The Sun makes 4,320,000 revolutions in a kalpa; the Moon 57,753,336; Mars 2,296,832; Mercury 17,937,060; Jupiter 364,220; Venus 7,022,377; and Saturn 146,568. These are the revolution-numbers of the planets in a kalpa, (which equal) the civil days. The last civil day is called the *dhruva* (fixed point); from this the motions of the planets begin in the kalpa.”

6.2 Epoch and Ahargana Calculation

कल्पपराद्ध मनवः षट्काश्य गताश्चतुर्द्वियुगानि युगानां । श्रुतपुराणादीनिक्षेपानां क्षयः सूर्यदिनं चक्षुः ॥
 नवगणराशिमुनिन् नव समन्वन्तरदेवः शकुन्तपाते । शर्वमतेः सम्बत्सरात् संख्यां शरवच्छतं रूपैः ॥

BrSpSi. I. 26–27

These verses give the epoch calculation. At the beginning of Brahmagupta’s *sr̥ṣṭi-saṁvatsara* (creation year), six manus have elapsed, along with 27 caturyugis of the seventh manu, and of the 28th caturyugi, three yugas (*kṛta*, *tretā*, *dvāpara*) plus 3179 years of *kaliyuga*.

6.2.1 Sanskrit Text on Trigonometric Methods

Brahmagupta’s rule for computing Rsines (BrSpSi. II. 15–16):

पूर्वाच्छिद्रच्छेदैः पूर्वसिद्धैर्ज्याविश्लेषैः सैषा । वर्गीकृत्या युक्ता त्रिज्यावर्गात् पदं शुद्धिः स्यात् ॥ १५ ॥
 तत्पदं चापकर्णस्त्रिज्यावर्गाद् विशोध्य शेषात् । पदं द्विघ्नं भुजज्याख्यं कोटिज्याख्या च सा समे ॥ १६ ॥

Translation: “The differences of the Rsines, already obtained successively, are each multiplied by the remaining portion of the chord and divided by the first Rsine. The square of this, subtracted from the square of the radius, gives the square root which is the *doḥ* (chord). The square root of the remainder, when the square of the radius is diminished by this, is the *kotijyā* (complement sine). In the (first) quadrant, these are respectively called the *bhujā-jyā* and *kotijyā*.”

And the rule for the shadow (BrSpSi. III. 7):

शङ्कुवर्गेण युक्तं दृग्वर्गं मूलं तु तत्कर्णः । शङ्गुना त्रिज्या हृता दृग्ज्या छायाया त्रिजया हृता ॥ ७ ॥

Translation: “The square of the *śaṅku* increased by the square of the *drg* (distance), the square root is the *karna* (hypotenuse). The *śaṅku* multiplied by the radius and divided by the shadow gives the *drgjyā* (Rsine of the zenith distance); (the *śaṅku* multiplied by the radius) divided by the shadow gives the corresponding (Rsine).”

Colophon

End of Chunk 3 (Pages 281–420)

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